Literature-Guided Minimax Optimization of Virtual Epilepsy Neurostimulation

Cathy Liu
Independent Research Project
https://github.com/liuzhitong330/tvb-llm-robust-neurostim

Abstract. Computational models of epilepsy promise patient-specific treatment design, but most optimization workflows still search for parameters that perform well on average. In neuromodulation, this is a weak target: a protocol that improves the mean response can still fail in the patient whose network is least tolerant to stimulation. We present a literature-guided minimax pipeline that couples PubMed-scale hypothesis extraction, The Virtual Brain (TVB) Epileptor simulations, and large-language-model-guided black-box optimization. The optimizer proposes either intrinsic model-control parameters or clinically interpretable external-stimulation protocols; TVB evaluates each proposal across sampled virtual patients; and the objective maximizes worst-case reward, defined as the negative variance of simulated seizure activity. In the intrinsic model-control experiment, the best archived parameter set improved worst-case reward from -0.5285 to -0.3182, a 39.8% gain over baseline, and maintained large gains in held-out and stress-test virtual cohorts. The clinical-style external-stimulation search produced a much smaller worst-case improvement (1.7%), and a 20-patient virtual cohort showed no aggregate benefit of the selected right-hippocampal stimulation protocol ($p = 0.9019$), despite a 55% responder rate and a positive temporal-lobe subgroup signal. These results identify a sharp distinction between what can be achieved by controlling model excitability directly and what is recovered by a constrained stimulation protocol. The study should therefore be read as an in silico proof of concept for robust, literature-aware neurostimulation design, not as clinical evidence.

1 Introduction

Drug-resistant epilepsy is a systems problem as much as a focal pathology. Seizures arise from nonlinear interactions among local excitability, long-range structural connectivity, and patient-specific seizure onset networks. Responsive neurostimulation and deep brain stimulation have established that electrical intervention can reduce seizure burden in selected patients, but their efficacy remains heterogeneous: electrode location, stimulation amplitude, disease subtype, and latent brain state all modulate response. A computational optimizer for epilepsy treatment therefore faces a demanding target. It must not merely improve a representative patient; it must avoid catastrophic failure in the patient whose network lies at the edge of the sampled distribution.

Virtual brain models provide a principled way to study this problem before clinical deployment. The Virtual Brain (TVB) platform simulates whole-brain neural mass dynamics constrained by empirical connectomes, while the Epileptor model captures seizure initiation, propagation, and termination using a compact dynamical-system representation [1, 2, 3]. These tools make it possible to evaluate candidate interventions across virtual cohorts, but they do not by themselves solve the search problem. A realistic intervention space contains continuous parameters, discrete targets, mechanistic priors from the literature, and strong safety constraints. Exhaustive search is expensive; naive average-case optimization is clinically fragile.

This project asks whether a large language model (LLM) can act as a literature-aware proposal engine inside a rigorous simulation loop. The LLM does not evaluate outcomes and is not treated as a scientific oracle. Instead, it proposes candidate parameters, grounded in a corpus of epilepsy and neuromodulation abstracts. TVB supplies the reward. The optimization criterion is minimax robustness: choose the protocol whose worst virtual-patient outcome is best.

The central result is intentionally bifurcated. When the optimizer is allowed to tune intrinsic model parameters controlling epileptogenicity and coupling, it finds a strong robust solution: worst-case reward improves by 39.8% in the archived optimization trajectory. When the same framework is restricted to a more clinically interpretable external-current boost at one brain region, improvement is much smaller and cohort-level effects are mixed. That contrast is scientifically useful. It suggests that the pipeline can discover robust

dynamical regimes, while also revealing how much harder it is to translate those regimes into a constrained stimulation protocol.

The contributions of this work are fourfold:

1. a reproducible research-to-simulation-to-optimization pipeline linking PubMed-scale literature mining, TVB Epileptor simulation, and LLM-guided minimax search;
2. a robust intrinsic-control result showing large worst-case improvement across virtual patients;
3. a translational stress test showing that external-stimulation gains are modest and patient-response heterogeneity is substantial; and
4. an open static research essay, codebase, and paper artifact designed for independent inspection and extension.

2 Background

2.1 Virtual epilepsy patients

Patient-specific computational epilepsy modeling has advanced from explanatory dynamical systems toward intervention planning. The Virtual Epileptic Patient framework combines structural connectivity, hypothesized epileptogenic zones, and neural mass models to simulate seizure spread and candidate interventions [3]. At the node level, the Epileptor model formalizes seizure onset and offset as transitions in a multi-timescale dynamical system [2]. These models are especially attractive for neurostimulation because they allow parameter perturbations to be tested repeatedly under controlled assumptions.

The major limitation is that the virtual patient is not a real patient. Simulation results inherit assumptions about connectivity, node dynamics, observation windows, and parameterization. This work therefore treats TVB as a hypothesis engine: a platform for asking whether an optimization idea is coherent and stress-testable, not a substitute for electrophysiology, imaging, or clinical trials.

2.2 Robust optimization for neuromodulation

Clinical neuromodulation is a worst-case-sensitive domain. A parameter set that helps many patients but worsens a minority may be unacceptable, especially when adverse responses are difficult to predict. Minimax optimization makes this concern explicit. Instead of optimizing the mean response, it maximizes the minimum reward across a sampled cohort. The cost is conservatism; the benefit is that failures are directly visible in the objective.

For expensive black-box simulations, Bayesian optimization is a common baseline because it uses a surrogate model to reduce the number of evaluations [7]. The premise here is complementary. Literature-aware LLM proposals may be useful when the search space contains named brain regions, mechanistic priors, and qualitative constraints that are awkward to encode in a standard kernel. The LLM is not trusted to be correct; it is used to generate informed guesses that the simulator can accept or reject.

3 Study Design

Figure 1 summarizes the pipeline. The study begins by mining epilepsy, digital-twin, and neurostimulation literature from PubMed. The extracted research gaps motivate a robust-transfer problem: can an intervention selected in simulation improve the worst virtual-patient outcome rather than merely the average outcome? Candidate interventions are then proposed by an LLM and evaluated by TVB. All reported numerical results in this manuscript are taken from checked-in repository artifacts; no new scientific result is introduced by the paper-generation script.

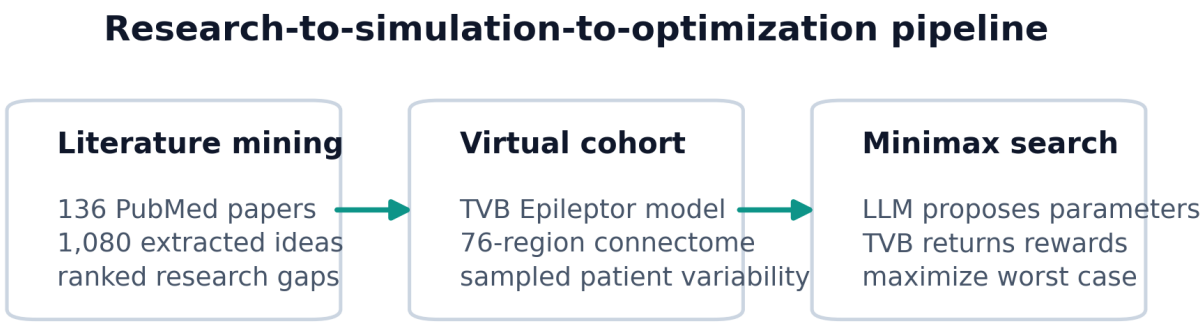

Figure 1: **Research-to-simulation-to-optimization pipeline.** A PubMed corpus is converted into structured research gaps, the highest-priority robust-transfer question is instantiated in a TVB Epileptor virtual cohort, and an LLM proposes candidate parameters that are accepted only after simulation-based reward evaluation. The objective is minimax: improve the least favorable sampled virtual-patient response.

4 Methods

4.1 Literature mining and hypothesis selection

The literature stage queried PubMed for work at the intersection of epilepsy, whole-brain modeling, stimulation, digital twins, and responsive neurostimulation. Deduplication was performed by exact PMID, yielding 136 unique papers. Titles and abstracts were passed to an LLM extraction prompt that returned structured open questions, future experiments, limitations, and TVB relevance. The final artifact contains 1,080 extracted ideas. A second LLM stage grouped ideas into thematic clusters and scored each cluster on novelty, TVB feasibility, and clinical impact. The robust-transfer study was selected because it scored highly on all three axes and could be tested directly in TVB.

This stage is a hypothesis-generation procedure. No claim is made that the extracted gaps are complete, deterministic, or expert validated. The role of literature mining is to bias search toward plausible questions; the role of simulation is to impose quantitative discipline.

4.2 Virtual brain model

Simulations use the TVB Epileptor model on a 76-region connectome loaded from the TVB library. For intrinsic-control experiments, candidate parameters are epileptogenicity x_0 and linear coupling K. The baseline configuration is $x_0 = -1.6$ and $K = 0.0152$. The optimized intrinsic setting found in the archived trajectory is $x_0 = -2.1$ and $K = 0.0165$.

For clinical-style external-stimulation experiments, the model holds intrinsic parameters fixed and applies an external-current boost b to one selected region. The candidate protocol is therefore $\theta = (b, s)$, where $b \in [0, 4]$ and $s \in \{0, \ldots, 75\}$ is a brain-region index. The best archived clinical-style recommendation is a low-intensity boost $b = 0.6$ at the right hippocampus (rHC, index 9).

4.3 Reward function

Let $x_{1,v}(t; \theta, p)$ denote the simulated fast Epileptor state at region v and time t for intervention parameters θ and virtual patient p. The seizure-burden score used by the code is the mean temporal variance across regions:

$$S(\theta; p) = \frac{1}{|V|} \sum_{v \in V} \mathrm{Var}_t \left[x_{1,v}(t; \theta, p) \right]. \tag{1}$$

The reward is its negative,

$$R(\theta; p) = -S(\theta; p), \tag{2}$$

so higher reward corresponds to lower simulated seizure-intensity variance. This is a model endpoint, not a clinical seizure-frequency endpoint.

4.4 Minimax objective

For a sampled cohort $\mathcal{P}$ of virtual patients, the robust objective is

$$\theta^\star = \arg\max_{\theta \in \Theta} \min_{p \in \mathcal{P}} R(\theta; p). \tag{3}$$

The training optimization used five virtual patients generated by uniform coupling perturbations with random seed 42. Additional archived artifacts evaluate the selected intrinsic parameters on held-out patients, wider perturbation regimes, and a 30-patient stress distribution.

4.5 LLM-guided search

The LLM search loop is deliberately simple. At each iteration, the model receives the prior optimization history and is asked to return a JSON object containing the next candidate parameters. TVB evaluates the candidate and appends the observed worst-case and mean rewards to the history. The intrinsic-control search tunes (x_0, K); the clinical-style search tunes (b, s). In the comparison artifact, a Gaussian-process Bayesian optimizer is run for the same number of external-stimulation evaluations and compared against the LLM-guided trajectory.

4.6 Virtual cohort analysis

The 20-patient cohort artifact compares no stimulation with the selected clinical-style rHC protocol ($b = 0.6$). Virtual seizure onset zone (SOZ) classes are sampled among hippocampal, frontal, temporal, and occipital categories. The analysis reports paired rewards, a paired t statistic, a two-sided p value, a standardized effect size, and responder/paradoxical-response rates. Because the cohort is simulated and small, these statistics are descriptive rather than inferential evidence for clinical use.

5 Results

5.1 Intrinsic minimax control produced a large robust improvement

The intrinsic optimization rapidly moved away from the baseline epileptogenicity and coupling regime. The best archived iteration occurred at $x_0 = -2.1$ and $K = 0.0165$, improving worst-case reward from -0.5285 to -0.3182 (39.8%) and mean reward from -0.4604 to -0.2736 (Figure 2A). The search trajectory is notable because several later proposals remained close to the same region of parameter space, suggesting that the simulator repeatedly rewarded a lower-epileptogenicity, slightly higher-coupling regime rather than a one-off artifact.

Held-out evaluation preserved the effect direction. In the archived test cohort, worst-case reward improved from -0.4985 to -0.3346, while mean reward improved from -0.4602 to -0.2848 (Table 1). The 30-patient stress distribution showed the same qualitative pattern: optimized worst-case reward was -0.4047 versus -0.5175 for baseline, and optimized mean reward was -0.3055 versus -0.4451.

5.2 Constrained external stimulation produced modest gains

The external-stimulation search was harder. The best archived protocol was low-intensity rHC stimulation ($b = 0.6$), improving worst-case reward from -0.5285 to -0.5194, or 1.7% (Figure 3A). This is directionally positive but much smaller than the intrinsic-control result. The difference is expected: intrinsic control directly changes the dynamical parameters that govern seizure propensity, whereas external stimulation perturbs one region in an already coupled system.

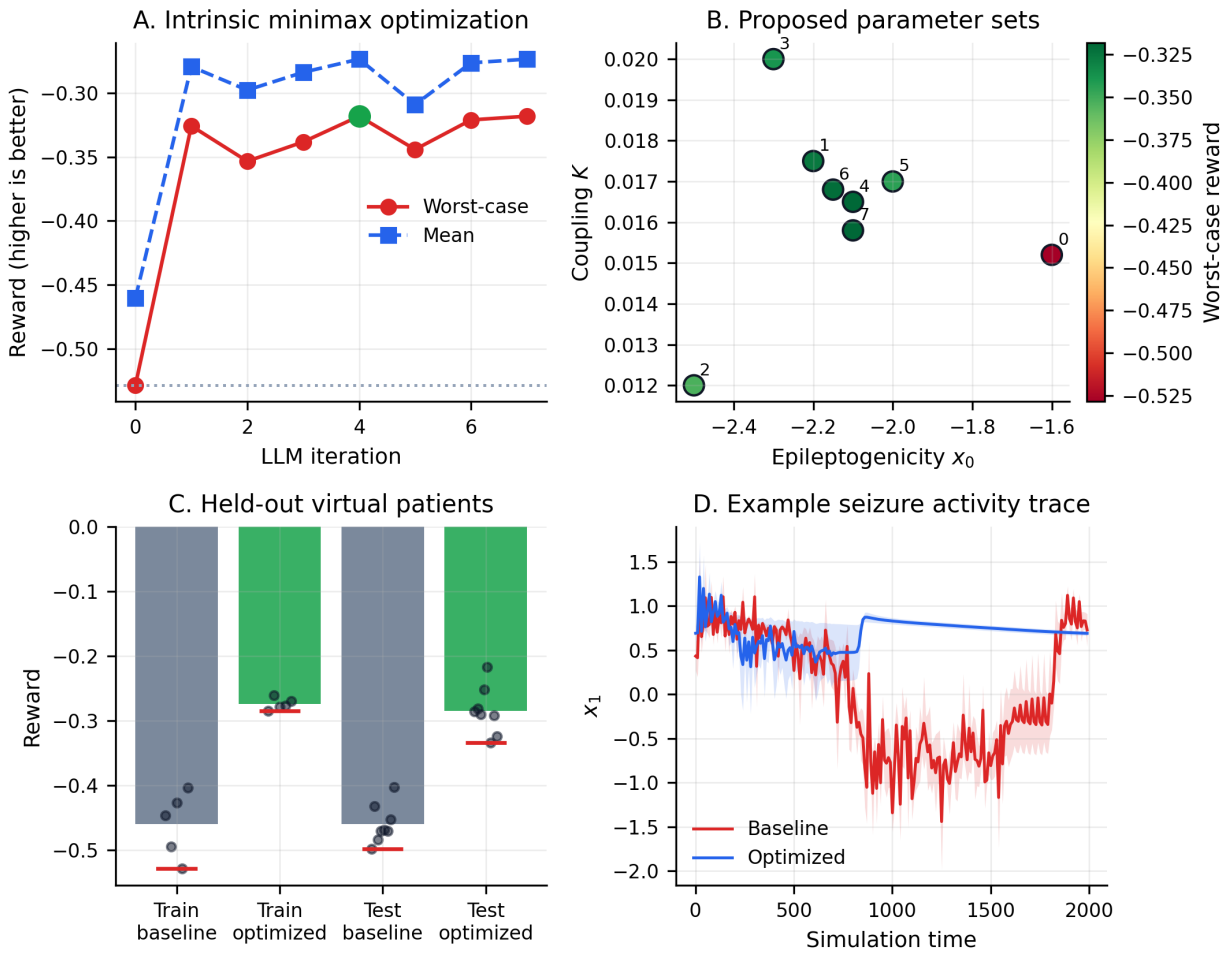

Figure 2: **Intrinsic minimax optimization. A**, LLM-guided optimization of epileptogenicity and coupling improved worst-case and mean reward relative to baseline. **B**, Proposed parameter sets concentrate around a lower-x_0 regime with slightly adjusted coupling. **C**, The optimized setting improves reward distributions in both training and held-out virtual patients; red horizontal marks denote worst-case rewards. **D**, Example simulated activity traces show lower mean fast-state variance under the optimized intrinsic regime.

The comparison with Bayesian optimization is informative but not definitive. Over eight evaluations, the LLM-guided external-stimulation trajectory reached a best reward of -0.5194, whereas the Bayesian optimizer reached -0.5221 (Figure 3B). The absolute margin is small (0.0027 reward units), but the LLM trajectory selected the literature-plausible rHC target immediately and improved from there. This supports the narrower claim that language-model priors can reduce wasted early evaluations in a structured, named-parameter search space.

5.3 The 20-patient cohort exposed responder heterogeneity

The 20-patient cohort is the most important translational check. Mean reward was -0.4501 at baseline and -0.4518 under the rHC protocol. The paired test was not significant ($t = -0.125$, $p = 0.9019$), and the standardized effect size was negligible ($d = -0.049$). However, the aggregate null result hides strong patient-level heterogeneity: 11 of 20 virtual patients improved, while 9 worsened.

SOZ stratification suggests one possible explanation (Table 2). The temporal subgroup had a positive mean reward change of $+0.0563$, while hippocampal, frontal, and occipital subgroups were near zero or negative. The temporal subgroup contains only two patients, so this should not be overinterpreted. Still,

Table 1: **Archived quantitative results.** Rewards are negative seizure-burden scores, so less negative values are better. Percent change is computed on worst-case reward unless otherwise stated.

Experiment	Condition	n	Worst reward	Mean reward	Worst-case gain
Intrinsic trace	Baseline	5	−0.5285	−0.4604	–
Intrinsic trace	Best LLM proposal	5	−0.3182	−0.2736	39.8%
Held-out intrinsic	Baseline	8	−0.4985	−0.4602	–
Held-out intrinsic	Optimized	8	−0.3346	−0.2848	32.9%
Stress intrinsic	Baseline	30	−0.5175	−0.4451	–
Stress intrinsic	Optimized	30	−0.4047	−0.3055	21.8%
External stimulation	No stimulation	5	−0.5285	–	–
External stimulation	rHC, $b = 0.6$	5	−0.5194	–	1.7%

Table 2: **SOZ-stratified virtual cohort response to rHC stimulation.** Values are optimized minus baseline reward. Positive values indicate lower simulated seizure burden under stimulation.

SOZ class	Patients	Mean reward change	SEM
Temporal	2	+0.0563	0.0297
Hippocampal	9	−0.0078	0.0191
Frontal	4	−0.0043	0.0302
Occipital	5	−0.0119	0.0363

the pattern is biologically plausible enough to motivate follow-up: stimulation at a hippocampal/temporal target may help a subset of temporal-network patients while being neutral or adverse elsewhere.

5.4 Network topology cautions against simple hub-targeting

The topology artifact adds a useful negative result. The best clinical-style target, rHC, is not a network hub in the loaded connectome: its degree is 1 and strength is 2.0. By contrast, prefrontal hubs such as rPFCORB have degree 32 and strength 71.0. A pure hub-targeting heuristic would therefore have avoided rHC. Yet the archived LLM and RAG trajectories repeatedly returned to hippocampal stimulation because of the epilepsy literature prior and observed rewards (Figure 4). This tension is important. Effective targets may be mechanistically local, topologically central, or state-dependent; whole-brain stimulation design should test all three hypotheses rather than assume that graph hubs are automatically optimal.

The RAG-augmented search did not outperform the simpler LLM-guided clinical search in the archived results. Its best reward was −0.5206, a 1.5% gain over baseline, and the best proposal appeared at the first iteration. Retrieval added interpretability by surfacing papers and mechanistic rationales, but it did not by itself solve the constrained optimization problem.

6 Discussion

This study demonstrates that literature-guided minimax search can identify robust dynamical regimes in virtual epilepsy models. The intrinsic-control result is the clearest evidence: across the archived optimization trace, held-out evaluation, and stress distribution, the selected parameter regime consistently reduces the simulated seizure-burden score. Because the objective is worst-case reward, the result is stronger than an average-case improvement: it explicitly improves the least favorable sampled virtual patient.

The translational story is more constrained. External stimulation at a single region produced only modest gains, and the 20-patient cohort did not show an aggregate benefit. This should not be read as a failure of the pipeline. It is precisely the kind of stress test a robust neurostimulation workflow should expose. The method found that direct model-control parameters can produce a large effect, then showed that the currently tested clinical proxy captures only a small portion of that effect and may worsen some virtual patients. In a clinical design setting, that information is valuable because it redirects attention toward patient selection,

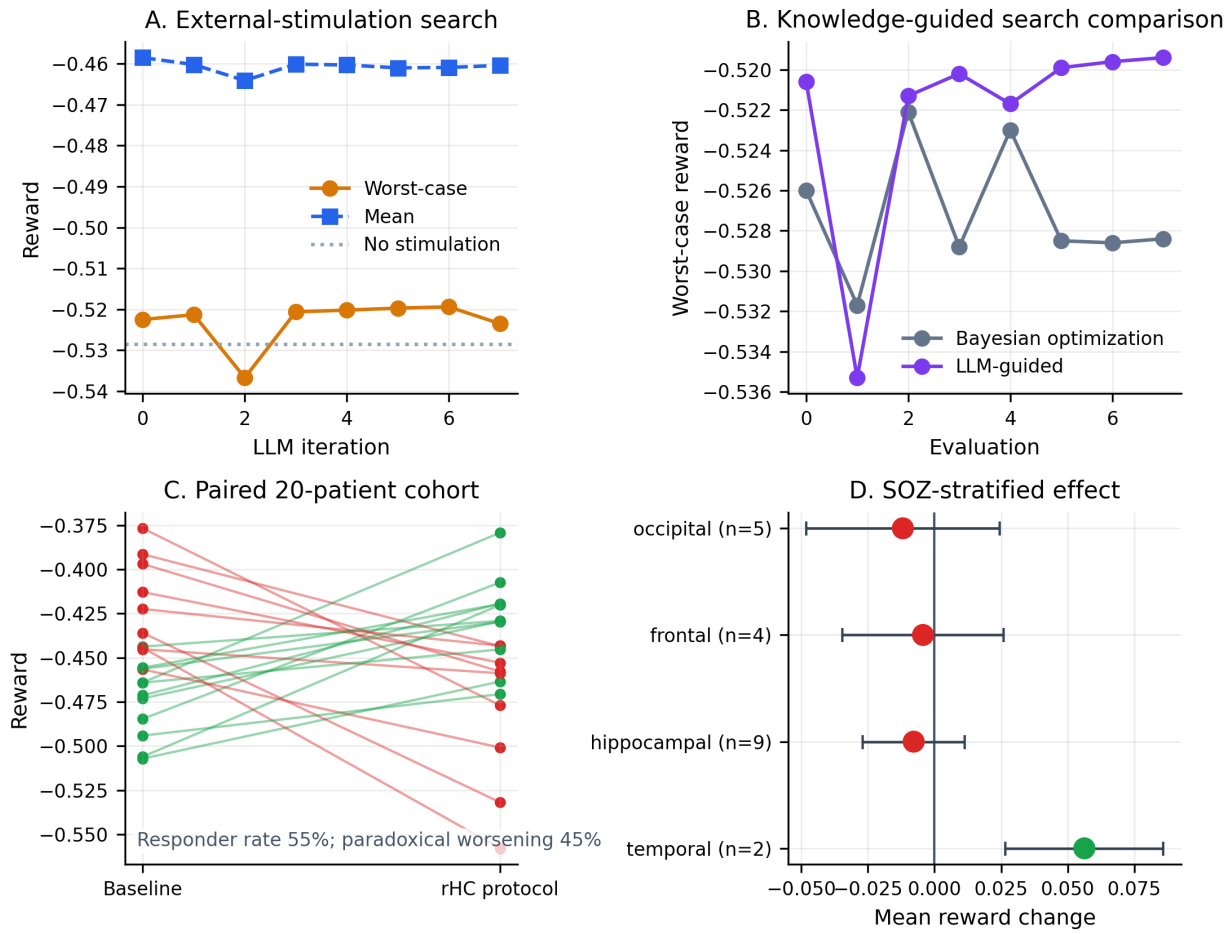

Figure 3: **Clinical-style stimulation and cohort response.** **A**, External-current stimulation produced only modest worst-case improvement relative to no stimulation. **B**, The LLM-guided trajectory slightly outperformed the Bayesian-optimization trajectory in the archived eight-evaluation comparison. **C**, In a 20-patient virtual cohort, the selected rHC protocol produced heterogeneous paired responses rather than a clear aggregate shift. **D**, SOZ-stratified analysis suggests a positive temporal subgroup signal but no consistent benefit across hippocampal, frontal, or occipital groups.

target stratification, waveform design, and closed-loop timing.

The results also clarify the role of LLMs in computational neuroscience. The LLM is useful as a proposal engine and literature interface, not as a source of truth. It can rapidly suggest plausible targets such as hippocampal stimulation for temporal-lobe epilepsy, explain why a candidate is mechanistically reasonable, and adapt proposals based on previous rewards. But every scientific claim in this workflow still depends on simulation output, artifact inspection, and domain validation. The architecture is therefore best understood as a constrained human-readable optimizer wrapped around a conventional computational model.

6.1 Limitations

Several limitations are substantial. First, the virtual cohort is small, and the patient variability model is simple. Coupling noise and synthetic SOZ categories are not substitutes for patient-specific diffusion MRI, stereo-EEG, seizure diaries, medication history, or implanted-device recordings. Second, the reward function is a proxy: negative variance of the simulated fast Epileptor state is useful for computational comparison but is not a validated endpoint for seizure frequency, severity, cognition, or quality of life. Third, the intrinsic-control result is not directly clinically actionable because changing x_0 and global coupling is a

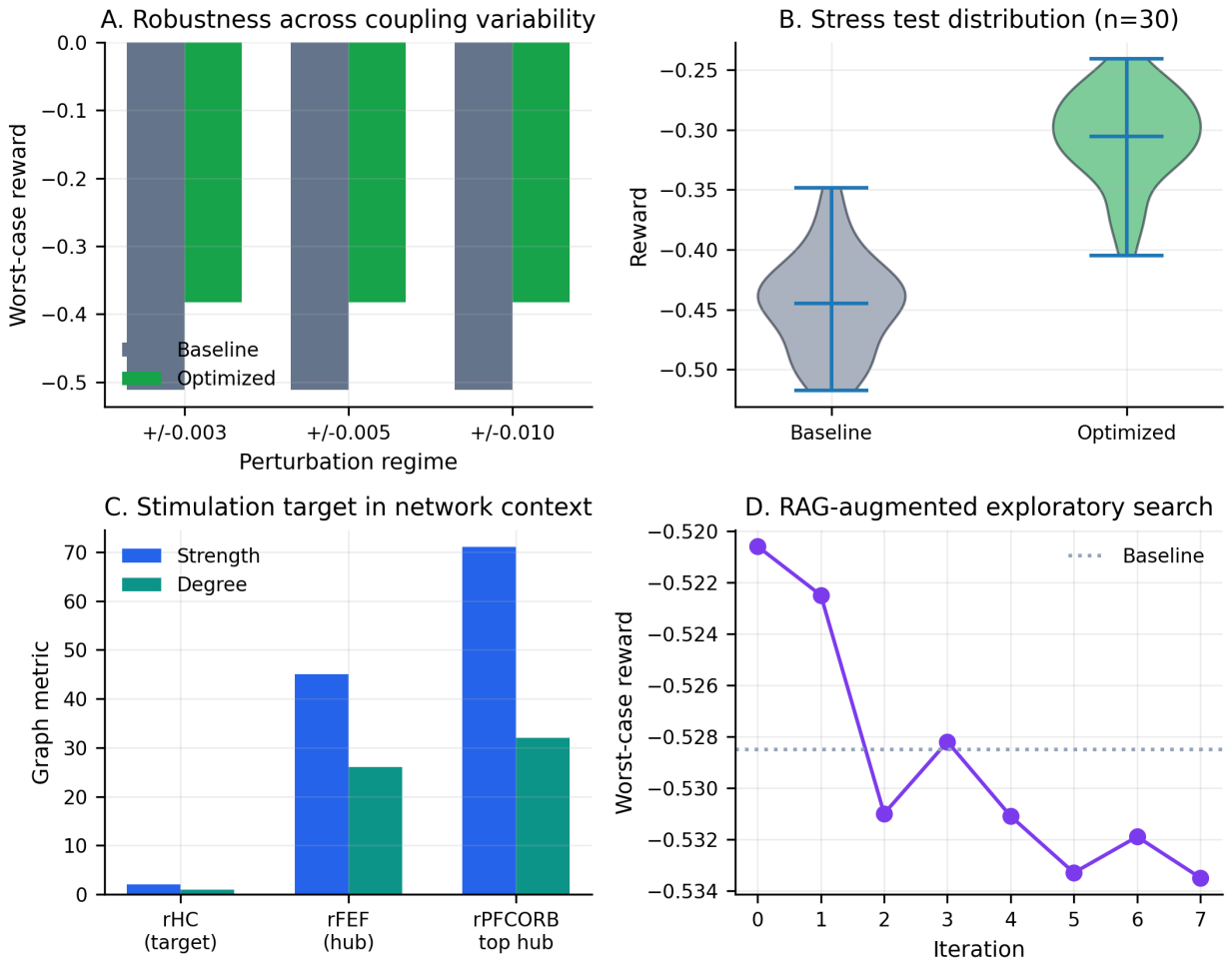

Figure 4: **Robustness, topology, and retrieval-augmented search. A**, The optimized intrinsic regime preserves worst-case improvement across archived perturbation regimes. **B**, Stress-test reward distributions remain shifted toward improved reward under optimized intrinsic parameters. **C**, The selected rHC stimulation target is far less connected than prefrontal hubs, cautioning against simple graph-hub heuristics. **D**, RAG-augmented search provided rationales and citations but did not exceed the best simpler LLM-guided external-stimulation result.

model intervention, not a neurosurgical protocol. Fourth, the literature-mining pipeline uses LLM-generated extraction, clustering, and scoring without expert adjudication. Its outputs should be treated as structured hypotheses. Fifth, the external-stimulation model abstracts away pulse width, frequency, timing, electrode geometry, tissue conductivity, stimulation artifacts, and safety constraints that dominate real devices.

6.2 Future work

The next version of this project should move from sampled virtual cohorts to patient-specific digital twins, with structural connectomes, seizure onset hypotheses, and validation against held-out electrophysiology. The stimulation model should include waveform, frequency, pulse width, phase, and closed-loop timing, because the external-current boost used here is too coarse to represent modern responsive neurostimulation. The optimization layer should compare LLM-guided proposals against stronger baselines, including random search, evolutionary strategies, constrained Bayesian optimization, and expert-designed protocols. Finally, the literature-mining stage should be reviewed by epilepsy clinicians and computational neuroscientists before it is used to prioritize translational experiments.

7 Conclusion

A robust neurostimulation optimizer should optimize for the patient who fails first, not the average patient who improves most easily. This project operationalizes that principle by coupling literature-guided LLM proposals with TVB Epileptor minimax evaluation. The intrinsic model-control experiment produced a large, reproducible in silico improvement in worst-case reward, while the clinical-style stimulation experiments exposed the expected difficulty of translating that control signal into a single-region protocol. The main contribution is therefore not a claim of clinical efficacy, but a disciplined framework for turning literature-derived hypotheses into stress-tested virtual interventions.

Code and Data Availability

All code, static result artifacts, figures, and the public GitHub Pages essay are available at https://github.com/liuzhitong330/tvb-llm-robust-neurostim. The figures in this manuscript are generated from checked-in JSON artifacts by `paper/generate_figures.py`. The paper does not introduce new simulation results beyond those artifacts.

Ethics and Clinical Use Statement

This work uses simulated virtual patients and contains no human-subject data. It is not medical advice, clinical guidance, or evidence that any stimulation protocol is safe or effective in patients.

Acknowledgments

This manuscript describes Cathy Liu's public research project on LLM-guided robust optimization for epilepsy neurostimulation using The Virtual Brain.

References

[1] P. Sanz Leon, S. A. Knock, M. M. Woodman, L. Domide, J. Mersmann, A. R. McIntosh, and V. Jirsa. The Virtual Brain: a simulator of primate brain network dynamics. *Frontiers in Neuroinformatics*, 7:10, 2013.

[2] V. K. Jirsa, W. C. Stacey, P. P. Quilichini, A. I. Ivanov, and C. Bernard. On the nature of seizure dynamics. *Brain*, 137(8):2210–2230, 2014.

[3] T. Proix, V. K. Jirsa, F. Bartolomei, M. Guye, and W. Truccolo. The Virtual Epileptic Patient: individualized whole-brain models of epilepsy spread. *NeuroImage*, 145:377–388, 2017.

[4] M. J. Morrell and the RNS System in Epilepsy Study Group. Responsive cortical stimulation for the treatment of medically intractable partial epilepsy. *Neurology*, 77(13):1295–1304, 2011.

[5] R. Fisher, V. Salanova, T. Witt, R. Worth, T. Henry, R. Gross, K. Oommen, I. Osorio, J. Nazzaro, D. Labar, and others. Electrical stimulation of the anterior nucleus of thalamus for treatment of refractory epilepsy. *Epilepsia*, 51(5):899–908, 2010.

[6] M. O. Baud, J. K. Kleen, E. A. Mirro, J. C. Andrechak, D. King-Stephens, E. F. Chang, and V. R. Rao. Multi-day rhythms modulate seizure risk in epilepsy. *Nature Communications*, 9:88, 2018.

[7] J. Snoek, H. Larochelle, and R. P. Adams. Practical Bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, 2012.